INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, NAGPUR
Department of Computer Science & Engineering
CSL471: Foundations of Generative AI(DSA/AIML)

Date: February 20, 2026

Sessional Exam
Duration: 1.5hour | Semester – VIII | Max. Marks: 30

Important Instructions:

- All the questions are compulsory.
- Assume suitable data wherever necessary

Q1a	An Energy-Based Model defines the energy function for classification as: $$E(x, y) = 0.5x_1y + 0.3x_2y + 0.4$$ Where, Input vector x=(4,2) and Possible class labels y ∈ {−1,+1} *Compute the energy values for both possible labels y= +1 and y=−1. *Based on the Energy-Based Model decision rule, determine the predicted class label.	(3) M	CO1
Q1b	An Energy-Based Model defines the conditional probability using the Boltzmann distribution: $$P(y|x) = \frac{e^{-E(x,y)}}{\sum_i e^{-E(x,y_i)}}$$ For a given input x, the energy values of two possible outputs are: • $E(x, y_1) = 0.5$ • $E(x, y_2) = 1.5$ (Use $e^{-0.5} = 0.6065$ and $e^{-1.5} = 0.2231$) i. Compute the partition function Z. ii. Calculate the probabilities P(y1|x) and P(y2|x). iii. Identify which output is more likely according to the Energy-Based Model.	(3) M	CO1
Q1c	A machine learning system uses two binary variables: • X: Input feature detected (1 = Yes, 0 = No) • Y: Correct prediction by an Energy-Based Model (1 = Correct, 0 = Incorrect) The joint probability distribution is given below: **X** 1 1 0 0 **Y** 1 0 1 0 **P(X,Y)** 0.3 0.2 0.1 0.4 i. Find the marginal probability P(X=1). ii. Compute the conditional probability P(Y=1|X=1). iii. Check whether X and Y are independent.	(4) M	CO1
Q2a)	A deep neural network consists of 5 hidden layers, each using the sigmoid activation function. During back propagation, the gradient passing through each layer is multiplied by the average derivative value of the sigmoid function. **Given:** Average sigmoid derivative at each layer: σ′(z)=0.2 and Gradient at the output layer: ∂L/∂a5=1	(2+2+1) M	CO2

	i. Calculate the gradient magnitude reaching the first hidden layer after back propagation through all 5 layers. ii. Analyze the computed result and determine whether the network experiences the vanishing gradient problem. Justify your answer mathematically. iii. Evaluate and state two pitfalls of the vanishing gradient problem affecting deep neural network training.		
Q2b	Explain the concept of compound scaling used in EfficientNet architecture. Illustrate your explanation by drawing a neat labeled diagram showing how network depth, width, and input resolution are scaled simultaneously. Using your diagram, analyze how compound scaling helps achieve better performance compared to scaling only one network dimension in traditional CNNs. State one architectural advantage and one limitation of EfficientNet design	(5) M	CO2
Q3a	Design a conceptual framework for a GAN based on minimax game theory and extends it to CycleGAN architecture by drawing a neat labeled diagram showing generators, discriminators, real data domains, and cycle-consistency flow. Using your designed framework, formulate the GAN minimax objective function and construct the cycle-consistency loss used in CycleGAN. Finally, propose one modification to improve training stability and justify how it overcomes a limitation of classical GAN training.	(5) M	CO2
Q3b	Critically evaluate the architecture and learning mechanism of a Boltzmann Machine by constructing a neat labeled diagram showing visible and hidden units with their interconnections. Based on your diagram, assess how energy minimization and stochastic neuron activation influence learning efficiency and model performance, and justify whether Boltzmann Machines remain practical for modern deep learning applications by identifying one key advantage and one significant limitation	(5) M	CO1

**********ALL THE BEST**********